

Karsten Kirby Uy

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Education

University of British Columbia

Bachelor of Applied Science in Computer Engineering (GPA: 88.8%)

Expected Graduation: Dec 2027

Vancouver, BC

- **Relevant Coursework:** Digital Systems Design (SystemVerilog), Operating Systems (C), Signals and Systems, Machine Learning with Engineering Applications (Python), Software Construction I (Java), Signals and Systems (DSP)

Experience

Intel

May 2026 – Present

System Simulation Engineer Intern

Vancouver, BC

- Developed **SystemC** transaction-level models of hardware blocks, enabling **pre-silicon** firmware and software validation ahead of RTL and silicon availability.
- Correlated model behavior against **RTL reference** simulations and integrated automated testbenches into a **CI regression** flow to catch model divergence early.

Cortyze

April 2026 – Present

Founding Engineer and CTO

Vancouver, BC

- Led the development of a cutting-edge AI-powered platform, driving innovation and growth in the company's core product line.

Synopsys Inc.

Jan 2025 – August 2025

Software Engineer Intern

Ottawa, ON

- Architected an **AI-driven data pipeline** in Bash and Python to process 20K+ samples, training ML models that achieved an **8x increase in verification coverage** and a **4.5x reduction in runtime**.
- Developed a **C++ feature extraction engine** to interface between hardware design files and ML models, automating the generation of complex TCL configuration scripts for distributed verification flows.
- Engineered cross-platform CLI and GUI workflows, streamlining 10+ compilation processes and ensuring deployment across **AlmaLinux and CentOS** environments using PyInstaller and C++ runtime integration.
- Built a comprehensive **automated regression suite** of 100+ hardware designs, utilizing performance profiling to quantify efficiency gains and ensure system stability across updates.

Insurance Corporation of British Columbia

May 2024 – Aug 2024

Software Developer Intern

North Vancouver, BC

- Developed a **three-tier full-stack web application** using **React, .NET, and Flask**, implementing role-based access control (RBAC) via Active Directory to replace a legacy system and centralize database management.
- Engineered a high-impact automation solution using **Python**, integrating with 5+ enterprise platforms to handle secure file transfers, significantly reducing annual operational costs and manual error rates.
- Resolved critical production incidents for 30+ enterprise solutions, maintaining high uptime through **proactive debugging, regression testing,** and the deployment of security patches.

Projects

UBC Room Finder | GitHub | Live Site | Next.js, PostgreSQL, Python, Selenium

- Designed, built, and maintained a production-ready full-stack web application for identifying real-time availability of UBC classrooms by aggregating timetable data into a fast, searchable system for on-demand room discovery.
- Developed and maintained a Python data ingestion pipeline using Selenium and BeautifulSoup to scrape 50+ pages and store 120,000+ UBC Online Timetable room booking records.
- Architected, deployed, and maintained a scalable full-stack system leveraging a PostgreSQL-backed Supabase database with custom RPCs and a responsive Next.js frontend (React, Tailwind CSS), delivering efficient room availability queries, dynamic filtering, and reliable production deployment via Vercel CI/CD.

FPGA Multi-Effects Audio Processor | GitHub | Demo | SystemVerilog, FPGA, Fixed-Point DSP, Python

- Engineered a synthesizable real-time audio DSP pipeline on the Intel Cyclone V (DE1-SoC): a 12-stage effect chain at 48 kHz/16-bit with sub-millisecond (<1ms) latency and an always-on YIN pitch tracker for chromatic tuning.

- Developed fixed-point DSP blocks including a 3rd-order polynomial $\tanh$ distortion model, an 8-sample lookahead compressor, and a multiplier-free Feedback Delay Network reverb of 8 fractional-delay lines coupled by a lossless 8×8 Hadamard matrix, holding 50 MHz timing closure through pipelining and multicycle-path constraints.
- Built a live JTAG-UART host link from an Avalon-MM adapter and a transport-agnostic command-parser FSM that lets a Python (Tkinter) GUI and CLI read, write, dump, and save a 512-byte parameter store on the running board.
- Architected power-persistent storage of 4 preset banks in EPCQ SPI flash via a custom erase/program FSM, with a gain-ramp fade FSM and GPIO/ADC pedal controls also mirrored to a class-compliant Arduino USB-MIDI controller.

Parallel ARC4 Decryptor | *SystemVerilog, FPGA, Cryptography*

Nov 2025 - Dec 2025

- Engineered a synthesizable ARC4 engine on DE1-SoC, instantiating 150+ parallel cores to achieve a throughput of 3,500+ keys/sec for ciphertext cracking.
- Achieved timing closure at 120 MHz by pipelining RTL logic, optimizing resource utilization (92%), and mitigating routing congestion during the Quartus fitting stage.
- Validated hardware performance and functional correctness using ModelSim and SystemVerilog UVM-inspired testbenches to verify throughput and parallel execution.

Moving Obstacle Detection and Navigation (MODaN) | *Python, ROS2, OpenCV, Kalman Filters* Jan 2026 – Apr 2026

- Engineered a real-time perception pipeline using an Intel RealSense D435i; implemented depth-gating and morphological filtering to maintain a deterministic 10Hz sample rate for obstacle segmentation.
- Developed a sensor fusion engine utilizing multi-frame Kalman filters to track dynamic obstacle trajectories, significantly reducing state estimation error caused by LiDAR noise and sensor occlusions.
- Architected a reactive navigation stack on the F1Tenth platform that "injects" 3D visual data into LiDAR-based Follow-the-Gap algorithms, enabling real-time autonomous course correction.
- Designed a ROS2 safety interlock node that evaluates Time-to-Collision (TTC) at predicted intercept points to preemptively override motor controller commands and prevent high-speed collisions.

Technical Skills

Languages: C++, C, Java, Python, JavaScript, C#, SQL, Bash, ARM Assembly, SystemVerilog, HTML/CSS

Libraries & Frameworks: React.js, Next.js, Tailwind CSS, Flask, SQLAlchemy, .NET, JUnit, Gradle, Selenium, OpenCV, JUCE, tkinter

Software & Platforms: Linux, AWS, GitHub, Perforce, MATLAB, Arduino, Visual Studio, Azure Data Studio, Vite, Blue Prism, OutSystems, ROS2

EDA Tools: VC Formal, ModelSim, Quartus

Electrical & Hardware: Multimeter, Oscilloscope, Soldering

Creative Tools: Studio One 5, Ableton, Audacity, FL Studio, DaVinci Resolve, Adobe After Effects, Adobe Premiere Pro, iMovie, CapCut

Productivity: Microsoft Excel, Word, Outlook, Teams, PowerPoint, OneNote